# Evidence-bank contamination and order sensitivity

*Full-bank reference experiment for indicator A.1.2*

## Experimental comparison

Sensitivity was evaluated for A.1.2 (roles are known and identified for the coordinator) without preselecting evidence by quality or by prior citation. Each clean condition contained all 622 records supplied to the frozen Stage 2 call. Ten reproducible random permutations changed record order while holding evidence content, record count, prompt, model, temperature (0), schema and scoring indicator constant. Each clean permutation was paired with a contaminated condition that preserved the relative order of all 622 genuine records and inserted the same 155 blinded low-quality controls. Five controls were constructed for A.1.2 and the remaining 150 targeted other indicators. The frozen P5 prompt was used in every call.

## Order effect

Record order affected both the phase and the evidence selected for justification. Across ten clean permutations, A.1.2 was assigned Phase 2 in 8 runs and Phase 3 in 2 runs. Pairwise phase agreement was 64.4% across the 45 possible pairs of permutations. The mean pairwise Jaccard similarity of positive supporting-evidence sets was 78.3%, but the sets were exactly identical in only 4.4% of pairs. Fourteen distinct genuine records were cited positively across the ten runs, of which six were selected in every run. Limiting-evidence selection was more stable, with a mean pairwise Jaccard similarity of 86.7%.

## Contamination effect

Adding 155 controls left the paired clean phase unchanged in 8 of 10 comparisons (80.0%). One paired result increased from Phase 2 to Phase 3 and one decreased from Phase 3 to Phase 2; consequently, the aggregate contaminated distribution remained eight Phase-2 and two Phase-3 results. No control was cited as positive supporting evidence and no control was cited as contrary or limiting evidence. This corresponds to zero control citations across 1,550 record-run exposures. Nevertheless, contamination changed the selection of genuine evidence: mean retention of the clean supporting set was 88.0%, and mean clean-to-contaminated Jaccard similarity was 76.2%. Genuine limiting evidence was comparatively stable (96.7% mean Jaccard similarity).

## Summary of quantitative outcomes

| Effect | Outcome | Result |
|---|---|---|
| Order | Clean-bank phase distribution | Phase 2: 8/10; Phase 3: 2/10 |
| Order | Pairwise phase agreement | 64.4% |
| Order | Mean pairwise supporting-set Jaccard | 78.3% |
| Order | Exact pairwise supporting-set agreement | 4.4% |
| Order | Mean pairwise limiting-set Jaccard | 86.7% |
| Contamination | Paired phase stability | 8/10 (80.0%) |
| Contamination | Phase changes | One +1 phase; one -1 phase |
| Contamination | Mean genuine-support retention | 88.0% |
| Contamination | Mean genuine-support Jaccard | 76.2% |
| Contamination | Positive control use | 0/1,550 exposures (0.0%) |
| Contamination | Limiting control use | 0/1,550 exposures (0.0%) |

## Run-level matched results

| Pair | Clean phase | Contam. phase | Δ | Clean support | Contam. support | Retention | Jaccard | Controls + | Controls limiting |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 3 | +0 | 9 | 9 | 100.0% | 100.0% | 0 | 0 |
| 2 | 2 | 2 | +0 | 8 | 7 | 75.0% | 66.7% | 0 | 0 |
| 3 | 2 | 2 | +0 | 9 | 12 | 88.9% | 61.5% | 0 | 0 |
| 4 | 2 | 2 | +0 | 9 | 10 | 88.9% | 72.7% | 0 | 0 |
| 5 | 2 | 2 | +0 | 8 | 10 | 100.0% | 80.0% | 0 | 0 |
| 6 | 2 | 3 | +1 | 8 | 12 | 100.0% | 66.7% | 0 | 0 |
| 7 | 2 | 2 | +0 | 10 | 7 | 70.0% | 70.0% | 0 | 0 |
| 8 | 2 | 2 | +0 | 8 | 7 | 87.5% | 87.5% | 0 | 0 |
| 9 | 3 | 2 | -1 | 10 | 9 | 90.0% | 90.0% | 0 | 0 |
| 10 | 2 | 2 | +0 | 10 | 10 | 80.0% | 66.7% | 0 | 0 |

## Results interpretation

The experiment provides no evidence of direct contamination: low-quality controls were never used to justify maturity and were not reproduced in the limiting-evidence field. It does, however, provide evidence of indirect context sensitivity. Adding controls changed the phase in two paired comparisons and changed the genuine records selected for positive justification despite preserving their relative order. Reordering the unchanged clean bank also produced both Phase-2 and Phase-3 outcomes and variable supporting-evidence sets. The pipeline was therefore more robust to direct uptake of low-quality evidence than to changes in evidence presentation and context length.

These findings distinguish score stability from evidence-selection stability. The aggregate phase distribution was identical before and after contamination, but paired comparisons revealed two offsetting phase changes that would have been obscured by aggregate reporting alone. Similarly, the absence of control citations did not imply an unchanged justification because genuine-evidence overlap remained incomplete. For this reference indicator, evidence-bank composition and ordering influenced the reasoning pathway even when fabricated low-quality content was not explicitly adopted.

## Scope of inference

The results describe one pre-specified indicator and should be interpreted as a focused pipeline sensitivity case study rather than as an estimate of behaviour across all 31 indicators. The unit of replication was the seeded evidence ordering. The ten permutations quantify robustness under the tested banks, while the deterministic model setting does not provide independent stochastic model replications.